
EXPENSE TRACKER APPLICATION

PROJECT DOCUMENTATION

Project Name : Expense Tracker Application

Project Type : Mobile Application

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Role : Flutter Developer

Technology Stack : Flutter
Dart
Firebase Authentication
Cloud Firestore
Material Design

Version : 1.0

Completion Date : 19 July 2026

Chapter 1: Introduction

1.1 Project Overview

The Expense Tracker Mobile Application is an Android application developed using Flutter with Firebase as the backend service. The application enables users to securely record, manage, update, search, analyze, and delete their daily expenses.

Each user creates an account using Firebase Authentication. After authentication, every expense is stored in Firebase Cloud Firestore under the authenticated user's account, ensuring that users can access only their own financial records.

The application follows a clean and simple user interface with a dark theme to provide a modern user experience. The project demonstrates the implementation of Flutter front-end development together with Firebase Authentication and Cloud Firestore integration.

1.2 Problem Statement

Many individuals struggle to maintain a record of their daily expenses. Traditional methods such as notebooks or manual calculations are time-consuming and prone to errors. Existing applications often include unnecessary complexity for users who simply want to record and monitor their spending.

This project addresses the problem by providing a lightweight mobile application that allows users to securely manage their expenses with essential features including adding, editing, deleting, searching, and viewing expense summaries.

1.3 Project Objectives

The primary objectives of the project are:

- To develop a secure expense management application.
- To implement Firebase Authentication for user login and registration.
- To store expense records using Cloud Firestore.
- To allow users to add new expenses.

- To allow users to edit existing expenses.
- To allow users to delete unwanted expenses.
- To provide expense searching by title or category.
- To display monthly and total spending summaries.
- To categorize expenses for basic analytics.
- To develop a responsive and user-friendly Flutter interface.

1.4 Scope of the Project

The application is designed for Android devices and focuses on personal expense management.

The project includes:

- User Registration
- User Login
- Password Reset
- Secure Authentication
- Add Expense
- Edit Expense
- Delete Expense
- Search Expenses
- Dashboard displaying Total Spending
- Monthly Spending Summary
- Expense Analytics by Category
- Cloud Storage using Firebase Firestore

The project does not include budgeting, notifications, export functionality, multi-currency support, or offline synchronization.

1.5 Development Tools and Technologies

Frontend

- Flutter
- Dart

Backend

- Firebase Authentication
- Firebase Cloud Firestore

Development Environment

- Android Studio
- Visual Studio Code

Version Control

- Git
- GitHub

1.6 Target Users

The application is intended for individuals who wish to maintain a digital record of their personal expenses.

The primary users include:

- Students
- Professionals
- Individuals managing personal finances

The application is designed for users with basic smartphone knowledge and does not require technical expertise.

Chapter 2: System Analysis

2.1 Existing System

Traditional expense tracking methods, such as maintaining handwritten records or spreadsheets, require manual calculations and organization. These methods are time-consuming, difficult to maintain, and increase the chances of errors or data loss.

Although many expense management applications are available, several include advanced features that may be unnecessary for users who only need a simple and efficient way to record daily expenses.

2.2 Proposed System

The proposed Expense Tracker Mobile Application provides a simple and secure solution for managing personal expenses.

The application allows users to create an account, log in securely, record expenses, modify existing records, delete unwanted expenses, search expenses by title or category, and view spending summaries.

Firebase Authentication ensures secure user access, while Cloud Firestore stores each user's expense records separately in the cloud.

2.3 Functional Requirements

The system provides the following functionality:

- User Registration
- User Login
- Password Reset
- User Logout
- Add Expense
- Edit Expense
- Delete Expense
- Search Expenses
- View Total Spending
- View Monthly Spending Summary
- View Category-wise Expense Analytics

2.4 Non-Functional Requirements

Performance

- The application should respond quickly to user actions.
- Expense records should be loaded in real time from Firebase Cloud Firestore.

Security

- User authentication is handled using Firebase Authentication.
- Each user can access only their own expense records.

Reliability

- Expense data is stored securely in Cloud Firestore.
- User data remains available after restarting the application.

Usability

- The interface should be simple and easy to navigate.
- Users should be able to perform common operations with minimal effort.

Maintainability

- The project follows a modular Flutter structure.
- Screens, models, and services are separated to improve maintainability.

Compatibility

- The application is designed for Android devices supporting Flutter.

2.5 System Modules

The application consists of the following modules:

Authentication Module

Responsible for user registration, login, password reset, and logout using Firebase Authentication.

Dashboard Module

Displays:

- Total Spending
- Monthly Spending
- Expenses grouped by date

Expense Management Module

Allows users to:

- Add expenses
- Edit expenses
- Delete expenses

Search Module

Allows users to search expenses using the expense title or category.

Analytics Module

Calculates and displays the total spending for each expense category.

Cloud Database Module

Stores and retrieves user data using Firebase Cloud Firestore.

2.6 Advantages of the Proposed System

- Secure user authentication.
- Cloud-based data storage.
- User-specific expense records.

- Easy expense management.
- Real-time synchronization with Firestore.
- Simple and clean user interface.
- Fast searching of expense records.
- Automatic calculation of spending summaries.

Chapter 3: System Design

3.1 System Architecture

The Expense Tracker Mobile Application follows a client-server architecture.

The Flutter application serves as the client-side interface, while Firebase provides backend services including Authentication and Cloud Firestore.

Architecture Components

- Flutter Mobile Application (Frontend)
- Firebase Authentication
- Cloud Firestore Database

The application communicates directly with Firebase services through the Firebase SDK. User authentication is handled by Firebase Authentication, and all expense records are stored in Cloud Firestore.

3.2 Application Workflow

The application performs the following sequence of operations:

1. User creates an account.
2. User logs into the application.
3. Firebase authenticates the user.
4. Dashboard is displayed.
5. User can:
 - a. Add Expense
 - b. Edit Expense
 - c. Delete Expense

- d. Search Expenses
 - e. View Analytics
6. Expense data is stored in Cloud Firestore.
 7. Dashboard automatically updates whenever the database changes.

3.3 Database Design

The application uses Cloud Firestore as its database.

Firestore Structure

users

└─ userId

└─ expenses

└─ expenseId

- id
- title
- amount
- category
- date

Each authenticated user has a separate collection of expense records, ensuring data isolation and security.

3.4 Expense Data Model

Each expense record contains the following attributes:

Field	Data Type	Description
id	String	Unique identifier of the expense

title	String	Expense title
amount	Double	Expense amount
category	String	Expense category
date	Timestamp	Date of the expense

3.5 User Interface Design

The application consists of the following screens:

Login Screen

Allows existing users to log into the application.

Features:

- Email
- Password
- Forgot Password
- Login Button
- Navigate to Sign Up

Signup Screen

Allows new users to create an account.

Features:

- Full Name
- Email
- Password
- Confirm Password
- Create Account Button

Dashboard Screen

Displays the user's expense overview.

Features:

- Total Spending
- Monthly Spending
- Expenses Grouped by Date
- Add Expense Button
- Search Button
- Analytics Button
- Logout Button

Add Expense Screen

Allows users to create a new expense.

Fields:

- Expense Title
- Amount
- Category
- Date

Edit Expense Screen

Allows users to modify an existing expense.

Users can update:

- Title
- Amount
- Category
- Date

Search Screen

Allows users to search expenses using:

- Expense Title
- Expense Category

Matching expenses are displayed instantly.

Analytics Screen

Displays total spending grouped according to expense category.

Example:

- Food
- Shopping
- Travel
- Bills

3.6 Navigation Flow

The application navigation follows the structure below:

Signup Screen

↓

Login Screen

↓

Dashboard

└─ Add Expense

└─ Edit Expense

└─ Search Expenses

└─ Analytics

└─ Logout

3.7 Security Design

The application uses Firebase Authentication to authenticate users.

Cloud Firestore stores expense data separately for every authenticated user using the following structure:

users/{userId}/expenses/{expenseId}

This ensures that users can only access and manage their own expense records.

3.8 Design Considerations

The following principles were followed during development:

- Simple and clean interface
- Dark theme for better visual comfort
- Modular Flutter project structure
- Secure cloud-based storage
- Responsive layouts
- Easy navigation between screens
- Separation of frontend and backend services

Chapter 4: Implementation

4.1 Development Environment

The Expense Tracker Mobile Application was developed using the following software and tools:

Component	Technology Used
Programming Language	Dart
Framework	Flutter
IDE	Visual Studio Code
Backend	Firebase
Authentication	Firebase Authentication

Database	Cloud Firestore
Operating System	Windows 11
Version Control	Git & GitHub

4.2 Project Structure

The project follows a modular folder structure to improve readability and maintainability.

lib/

|— main.dart

|

|— models/

| └─ expense_model.dart

|

|— screens/

| └─ login_screen.dart

| └─ signup_screen.dart

| └─ dashboard_screen.dart

| └─ add_expense_screen.dart

| └─ edit_expense_screen.dart

| └─ search_screen.dart

| └─ analytics_screen.dart

|

└─ services/

└─ auth_service.dart

Each file has a specific responsibility, making the application easier to maintain and extend.

4.3 Firebase Authentication Implementation

Firebase Authentication is used to manage user accounts securely.

The implemented authentication features include:

- User Registration
- User Login
- Password Reset
- User Logout

Each user must authenticate before accessing the dashboard.

4.4 Cloud Firestore Implementation

Cloud Firestore stores all expense records online.

Each authenticated user has a separate collection of expenses.

Firestore Structure:

users

└─ userId

└─ expenses

└─ expenseId

Each expense contains:

- Expense ID
- Title
- Amount
- Category
- Date

4.5 Expense Management Module

The application provides complete CRUD functionality.

Add Expense

Users can enter:

- Title
- Amount
- Category
- Date

The expense is saved to Cloud Firestore.

Edit Expense

Users can update:

- Title
- Amount
- Category
- Date

The updated information is stored immediately in Firestore.

Delete Expense

Users can permanently remove an expense.

A confirmation dialog is displayed before deletion.

4.6 Dashboard Implementation

The Dashboard acts as the application's home screen.

It displays:

- Total Spending
- Monthly Spending
- Expenses grouped by date

The dashboard uses Firestore StreamBuilder, allowing data to update automatically whenever changes occur.

4.7 Search Implementation

The search feature allows users to find expenses using:

- Expense Title
- Expense Category

Search results update instantly as the user types.

4.8 Analytics Implementation

The Analytics screen calculates the total amount spent for each expense category.

For each expense:

- Category names are collected.
- Amounts belonging to the same category are summed.
- Results are displayed in a categorized list.

4.9 User Interface

The application follows a consistent dark theme.

Common design elements include:

- Rounded input fields
- Material Design widgets
- Green accent color
- Responsive layouts
- Scrollable screens
- Simple navigation

4.10 Error Handling

Basic validation has been implemented throughout the application.

Examples include:

- Preventing empty input fields
- Password confirmation during signup
- Firebase authentication error messages
- Safe navigation after asynchronous operations using `mounted`

4.11 Summary

The Expense Tracker Mobile Application was successfully implemented using Flutter and Firebase.

The system supports secure user authentication, cloud-based storage, real-time expense management, searching, editing, deletion, and category-based expense analytics. The modular implementation improves readability, maintainability, and future scalability.

Chapter 5: Testing and Results

5.1 Testing Overview

The Expense Tracker Mobile Application was tested throughout the development process to ensure that all implemented features functioned correctly.

Testing was performed on an Android device using Flutter.

The objective of testing was to verify that:

- User authentication works correctly.
- Expense records are stored in Cloud Firestore.
- Data retrieval is accurate.
- Expense modification functions properly.
- Search returns correct results.
- Analytics display correct category totals.
- Dashboard updates automatically after database changes.

5.2 Functional Testing

Module	Test Performed	Result
User Registration	Created a new account	Passed
User Login	Logged in with registered account	Passed
Password Reset	Reset email sent successfully	Passed
Logout	User logged out successfully	Passed
Add Expense	Added a new expense	Passed
View Expenses	Displayed expenses on dashboard	Passed
Edit Expense	Updated expense information	Passed
Delete Expense	Deleted selected expense	Passed
Search Expense	Searched by title and category	Passed
Analytics	Displayed category-wise totals	Passed
Monthly Spending	Calculated current month expenses	Passed
Total Spending	Calculated total expenses	Passed

5.3 Test Results

User Authentication

The authentication module was tested using multiple user accounts.

Results:

- Account creation completed successfully.
- Registered users were able to log in.
- Invalid credentials generated appropriate error messages.
- Password reset emails were successfully sent.
- Users could log out without issues.

Expense Management

The expense management module was tested by creating, updating, and deleting multiple expense records.

Results:

- Expenses were successfully stored in Cloud Firestore.
- Updated expenses reflected immediately.
- Deleted expenses were removed from both Firestore and the dashboard.
- Expense data remained linked to the authenticated user.

Dashboard

The dashboard was tested after each CRUD operation.

Results:

- Total spending updated automatically.
- Monthly spending updated correctly.
- Expenses were grouped by date.
- Newly added expenses appeared immediately.

Search Module

Different search queries were tested.

Results:

- Expenses could be searched using the title.
- Expenses could be searched using the category.
- Matching results updated instantly as the user typed.

Analytics Module

Expenses from multiple categories were added for testing.

Results:

- Total amount for each category was calculated correctly.
- Categories with multiple expenses displayed the combined total.
- Categories without expenses were not displayed.

5.4 Overall Results

The implemented features operated successfully during testing.

The application successfully provides:

- Secure user authentication
- Cloud-based expense storage
- Real-time dashboard updates
- Expense editing and deletion
- Expense searching
- Category-wise expense analytics

No major functional issues were observed after integrating Firebase Authentication and Cloud Firestore.

5.5 Summary

Testing confirmed that all implemented modules performed according to the project requirements. The Expense Tracker Mobile Application successfully manages personal expenses while providing secure authentication, cloud storage, and real-time data synchronization.

Chapter 6: Conclusion and Future Work

6.1 Conclusion

The Expense Tracker Mobile Application was successfully designed and developed using Flutter and Firebase. The application provides users with a simple and secure platform for recording and managing daily expenses.

The system implements user authentication using Firebase Authentication, ensuring that each user's financial data remains private. Cloud Firestore is used to store expense records, allowing real-time data synchronization and reliable cloud storage.

The application includes the following completed features:

- User Registration
- User Login
- Password Reset
- Secure Logout
- Add Expense
- Edit Expense
- Delete Expense
- Search Expenses
- Dashboard with Total Spending
- Monthly Spending Summary
- Expense Grouping by Date
- Category-wise Expense Analytics

All implemented features were tested and functioned successfully. The project meets its primary objective of providing a lightweight and user-friendly expense management application.

The project also provided practical experience in:

- Flutter mobile application development
- Firebase Authentication
- Cloud Firestore database integration
- CRUD operations
- Real-time data handling
- UI design using Material Design principles

Overall, the application demonstrates the practical implementation of modern mobile application development technologies and fulfills the intended project requirements.

6.2 Future Work

Although the application successfully implements the planned features, several enhancements can be added in future versions.

Possible future improvements include:

- Expense budget management
- Monthly spending limits with notifications
- Graphical charts and visual reports
- Expense filtering by custom date range
- Export expenses to PDF or Excel
- Receipt image attachment
- Dark and Light theme switching
- Multiple currency support
- Offline data synchronization
- Home screen widgets
- Push notifications and reminders
- Backup and restore functionality

These enhancements would improve usability and provide users with a more comprehensive personal finance management experience.

6.3 Final Remarks

The Expense Tracker Mobile Application successfully demonstrates the integration of Flutter with Firebase services to build a secure, cloud-based mobile application. The project serves as a practical example of mobile software engineering principles and provides a strong foundation for future enhancements and real-world deployment.